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Restore-RWKV: Efficient and Effective Medical Image Restoration with RWKV

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七月份挂在 Arxiv 上的一篇文章。

Mamba 在 24 年可以说是非常的火，self-attention 基本上可以被 Mamba 的 ss2d 替换，目前 mamba 已经有一千多引用了，可以想象未来还有成千上万的论文来水这个，就像之前水 self-attention 一样。

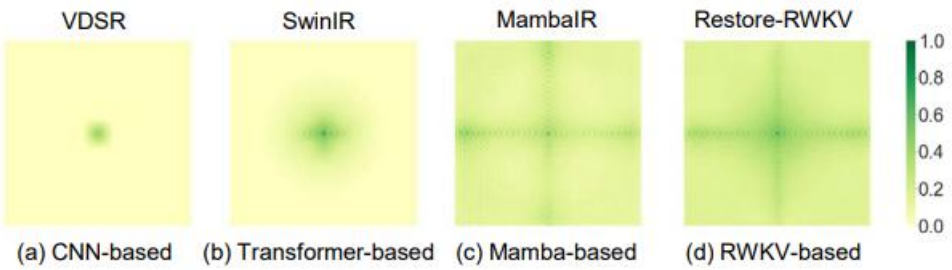


Figure 1. The effective receptive field (ERF) [27] visualization for different models. A more extensively distributed dark area indicates a larger ERF. Our proposed Restore-RWKV achieves the most significant global ERF.

Mamba、xLSTM 和 RWKV 的提出，主要还是用于解决 self-attention 二次方复杂度问题，更有效的进行全局建模。

Mamba 、xLSTM 和 RWKV 都是源于 NLP 任务，相对来说 xLSTM 跟 RWKV 热度没有 Mamba 那么大，不过，这对我们来说，是水的一个机会。

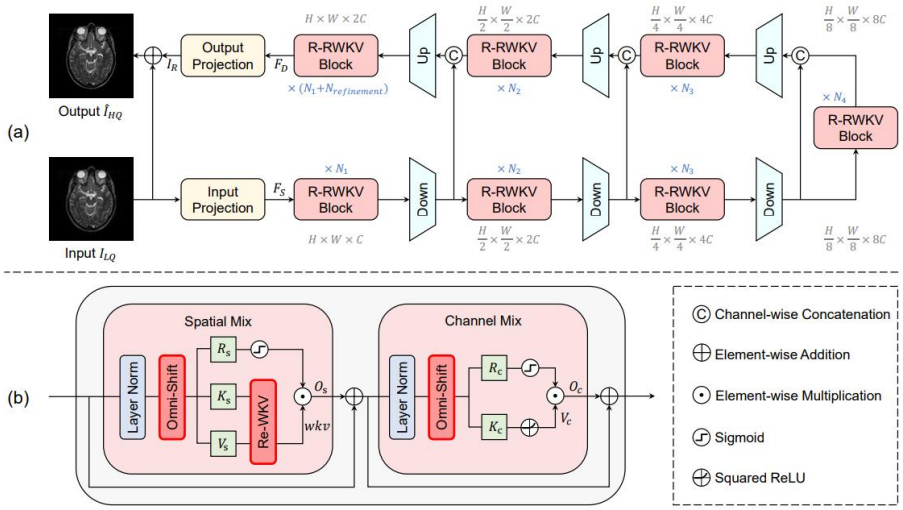


Figure 2. (a) Overview of the Restore-RWKV architecture. (b) Illustration of the R-RWKV block, which incorporates a Re-WKV attention mechanism to model global dependencies with linear complexity, and an Omni-Shift layer to capture local context.

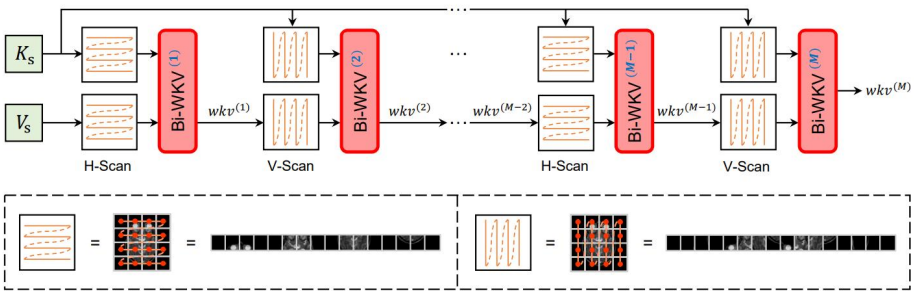


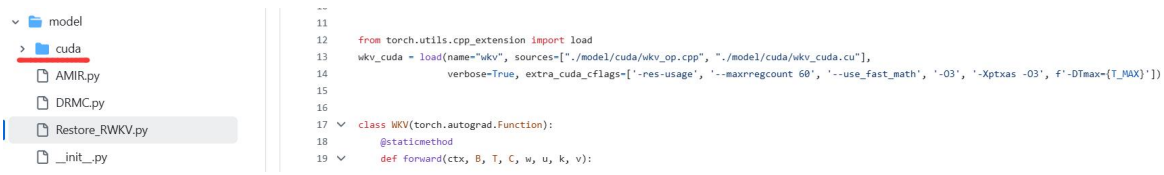
Figure 3. Illustrations of the Re-WKV attention mechanism. Re-WKV employs Bi-WKV [32] as its basic attention operator and applies Bi-WKV attention to 2D images recurrently through various scan directions to better model global dependencies.

整体结构还是沿用 Transformer 的架构，在 token mixer 和 ffn 中进行改进。在 token mixer 部分使用到 Re-WKV， Re-WKV 的目的是以线性复杂度实现 2D 图像全局建模，Omni-Shift 则用于捕获局部信息。

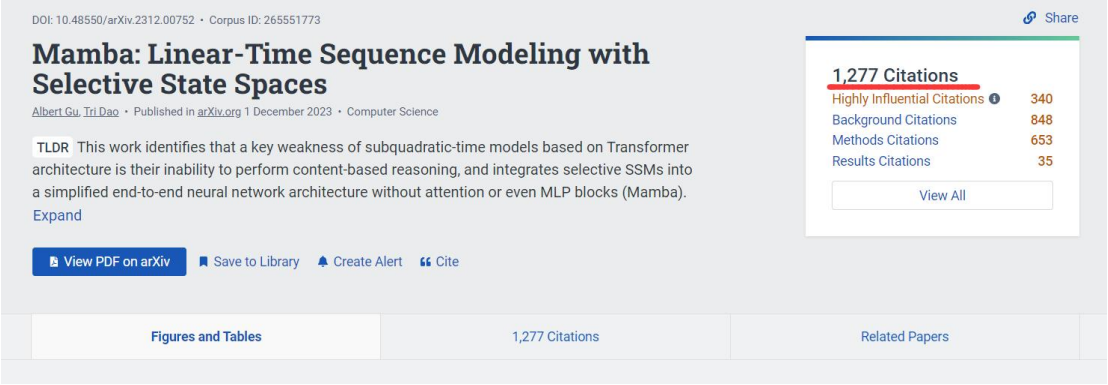
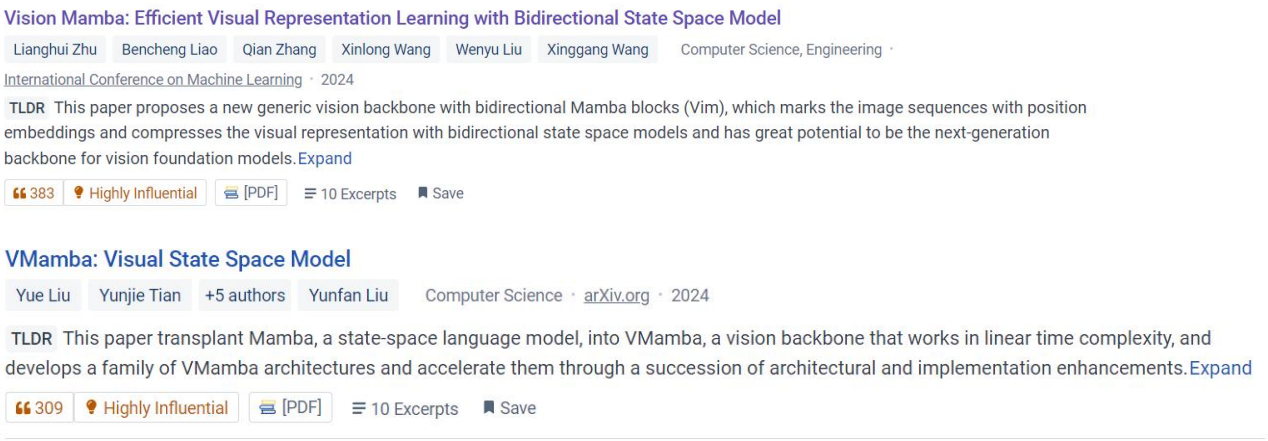
至于原理，大家可以先不用了解，可以先用一下在自己的任务中，试试效果，有提升，或者自己感兴趣，再去看看。

Github 代码地址：https://github.com/Yaziwel/Restore-RWKV

(任务）大家可以按照之前提取即插即用模块的流程，试着自己把这俩模块提取出来。



需要把这个包也下载下来。



PS：Mamba 引用已经有 1277 个了，相关顶会顶刊也有很多了。